

6

pivots and charts

Slice and Show

PivotTables, slicers, charts — and the killer feature called Refresh

Five days of pipeline, cleaning, merging, descriptive stats and regression. Today you turn yesterday's numbers into the *analytical layer* — the PivotTables and charts that answer Priya's questions ten different ways with three clicks each.

Today's tools: **PivotTables** for slicing the data without writing ten formulas. **PivotCharts** for visuals that update with the pivots. **Slicers** for interactivity. **The Chart Test** for picking the right chart every time.

And the bit that justifies everything you did in the first three days — **Refresh**. Press one button and the entire pipeline reruns against new source data. That's not a nice-to-have. That's the reason the whole Power Query architecture exists.

Friday already. Have we actually done it?

— you, looking at five days of work

Almost. Today is the part Priya sees.

the only day she'll remember.



BRAIN POWER

You built five days of work. Imagine next month, fresh source data arrives. How much of your work would you have to redo to get the analysis updated? Jot down your guess before turning the page. The Refresh feature on page 7 might surprise you.

turn over — into Excel, and into the pivots →

Friday afternoon, in Excel

Yesterday's numbers are on the sheet. Today we turn them into something Priya can actually *click around in*. And we build it so it auto-updates on Refresh.

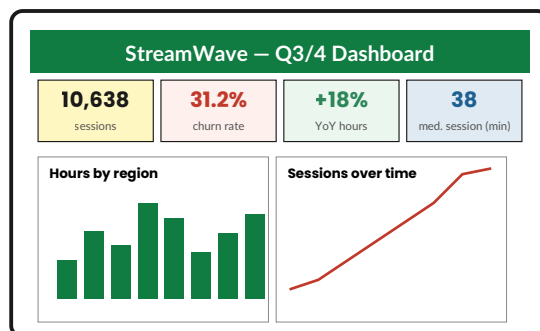
THE DELIVERABLE

- headline KPIs ✓
- 2 charts
- slicers (region, plan)
- conditional fmt
- methodology note

exec meeting 3pm

Priya:

"I just need three numbers and a story. Whatever's behind it can stay behind it."



DAY 6

Build a PivotTable

A PivotTable is the workhorse for analysis presentation. One click, four drop zones, and you can answer questions like "sessions per region per plan type" without writing a single formula.

Step 1 — Insert the PivotTable

1. Click anywhere inside the `Activity` Table you loaded on Day 4.
2. **Insert → PivotTable**. A dialog appears. Leave Source (Excel pre-filled `tblActivity`); set Destination to **New Worksheet**. Click OK.
3. A new sheet (rename it `Pivots`) opens with an empty pivot on the left and the *PivotTable Fields* pane on the right.

Step 2 — Drag fields into the four drop zones

The Fields pane has four drop zones at the bottom: **Rows** (left-hand labels), **Columns** (top headings), **Values** (the cell calculations), and **Filters** (a top-of-table dropdown). To answer "sessions per region per plan type":

- Drag `region` → **Rows**.
- Drag `plan_type` → **Columns**.
- Drag `record_id` → **Values** (Excel will say "Count of record_id").

You now have a small table: regions down the side, plans across the top, session counts in the cells.

[SCREENSHOT 1 — PIVOTTABLE FIELDS PANE]

The PivotTable Fields pane open on the right of an Excel worksheet, with the four drop zones populated as described. The resulting PivotTable visible on the left with region rows, plan_type columns, and Count of record_id in the cells.

SHOW THIS OFF

As you drop the third field, say: *"I just answered 'sessions by region by plan' in three drag-and-drops. Same answer would have been three nested SUMIFS. PivotTable is the right tool for cross-tabulation."*

Change values, group dates, sort

Change the aggregation

By default, PivotTables count text fields and sum numeric ones. To change it:

1. In the Fields pane, click the dropdown arrow next to the field in the **Values** box (e.g. "Count of record_id").
2. Click **Value Field Settings...**
3. Choose the function you want: Sum, Average, Max, Min, Count, Count Numbers, etc.
4. Click OK.

Example: swap `record_id` in Values for `duration`, set the function to **Sum**, then divide manually by 60 in a separate cell to convert to hours. Or use **Average** to get mean session length per region per plan.

Group dates by month or year

When you drop a date field into Rows or Columns, Excel often auto-groups it. If it doesn't, or if you want different grouping:

1. Right-click any date value in the PivotTable.
2. Click **Group...**
3. In the dialog, tick the level(s) you want: Days, Months, Quarters, Years. (Ticking multiple builds a hierarchy.)
4. Click OK. The dates now group into the chosen periods.

Sort largest first

Right-click any value cell → **Sort** → **Sort Largest to Smallest**. The pivot reorders by the value column you clicked. To filter to the top N: right-click a row label → **Filter** → **Top 10...** → set "Top" and the number you want.

SHOW THIS OFF

Whenever you change a Pivot, narrate what just happened. *"I changed the aggregation from Count to Sum, then divided by 60 – so this column is now total hours, not total minutes."* Pivots are powerful but invisible to the audience; saying what you did makes the audit trail audible.

Verify the pivot — Two-Route Check on totals

PivotTables fail silently. A miscounted filter, a hidden row, a Sum on a text-typed column — you can get a clean-looking pivot that's quietly wrong. Apply **The Two-Route Check** (the named device from Day 4) before you trust the headline number.

The check is simple: take any grand-total value in the pivot and reproduce it with a single SUMIFS or COUNTIFS against the raw table. If the pivot's grand-total cell reads *87 cancelled subscribers across all regions*, then `=COUNTIFS(tblSubs[status], "cancelled")` against the source table should also return 87. If the two numbers disagree by even one row, the pivot has a hidden filter, a wrong field type, or a stray group level — and you go fix it before it leaves your screen.

Do it once per pivot. Note the result in your methodology log. The assessor will ask "*how do you know that number is right?*" and you'll have an answer.

Slicers and timelines

A slicer is a clickable filter that sits next to your PivotTable. Priya clicks a button (e.g. "Premium") and the entire pivot – and any chart linked to it – updates instantly. No formulas, no faff.

Add a slicer

1. Click anywhere inside your PivotTable.
2. **PivotTable Analyze** → **Insert Slicer** (on the ribbon). A dialog lists every column in the source table.
3. Tick the columns you want as slicers – typically `region` and `plan_type`. Click OK.
4. Excel drops one slicer box per ticked column onto the sheet. Drag them where you want them.
5. Click any button on a slicer to filter. Hold Ctrl to multi-select. Click the filter-clear button (top-right of each slicer) to reset.

Add a timeline slicer for date columns

1. Click inside the PivotTable.
2. **PivotTable Analyze** → **Insert Timeline**. The dialog lists date columns from the source.
3. Tick `watch_date` → OK.
4. A horizontal timeline appears. Drag the handles to select a date range; click the unit dropdown (top-right) to switch between Days / Months / Quarters / Years.

Connect one slicer to multiple PivotTables

Right-click the slicer → **Report Connections...** → tick every PivotTable you want it to filter. Now one click on the slicer filters the whole dashboard.

ASSESSOR FAVOURITE

Slicers connected to multiple PivotTables is a distinction-band feature – it shows you understand the dashboard as a system, not just isolated tables. When you demonstrate, say: *"I'm connecting this slicer to both pivots so one click filters them together. That's what makes the dashboard interactive."*

PivotCharts

A PivotChart is the visual twin of a PivotTable. Click a slicer and both update together. That's what makes a dashboard feel alive.

Insert a PivotChart

1. Click anywhere inside your PivotTable.
2. **PivotTable Analyze** → **PivotChart**. The chart-type dialog appears (same as the regular chart dialog).
3. Pick the chart type — for "sessions by region", choose **Clustered Column**. Click OK.
4. Excel drops the chart onto the sheet. It's linked to the pivot — changes to either propagate to both.

Tidying the chart

- **Chart title**. Click on the auto-title, type the title you actually want. *"Sessions by Region – Q4 2024"*.
- **Axis labels**. Click the chart → **Chart Design** → **Add Chart Element** → **Axis Titles** → tick Primary Horizontal and Primary Vertical. Then click each to rename.
- **Field buttons**. The grey buttons Excel adds to a PivotChart (e.g. "region", "plan_type") are useful for you but clutter for Priya. To hide: right-click any field button → **Hide All Field Buttons on Chart**.

When to use a PivotChart vs a regular chart

PivotChart when the chart needs to follow the slicers. **Regular chart** when you want a fixed visual that doesn't change (e.g. a year-on-year headline chart that's always the same view). For most dashboards, mix both: PivotCharts for interactive sections, regular charts for the static headlines at the top.

SHOW THIS OFF

Demonstrate the interactivity live: click a slicer, watch both the PivotTable and the PivotChart update together. Say: *"One click filters everything connected to that slicer. That's why we built the dashboard this way – Priya can answer her own follow-up questions without me at the keyboard."*

Press Refresh

Yesterday's numbers, today's pivots, tomorrow's dashboard — all of it sits on top of the Power Query pipeline you built in the first three days. Which means when next month's data lands, you don't redo any of it. You press one button and the whole thing updates.

What Refresh actually does

Data → Refresh All (or Ctrl+Alt+F5). One click, in this order: Power Query re-reads every source file, the Cleaning Sequence reruns, the canonical key transformation reruns, merges and appends rerun, Tables on sheets update, every formula recalculates, every PivotTable rebuilds, every PivotChart redraws.

Total work for you: zero. Total time: usually under thirty seconds.

Why this is the killer feature

Without Power Query, when next month's data arrives you'd be opening files in Excel, copy-pasting, redoing your VLOOKUPs, fixing the dates, re-cleaning the genres, rebuilding the pivot, re-pointing the charts — a full afternoon at least, probably with a bug or two introduced. With Power Query, it's one click.

This is the actual business value of doing the upstream work properly. It's also the answer to one of the most common assessor questions: *"why did you use Power Query rather than just doing this in Excel?"*

ASSESSOR FAVOURITE

Asked *"what happens when next month's data arrives?"*, the one-sentence answer: *"I overwrite the source file in the Data folder, hit Data → Refresh All, and the entire pipeline reruns — cleaning, merges, formulas, pivots, charts — all in under thirty seconds. No rework."*

SHOW THIS OFF

Demonstrate Refresh live if you can. Make a small change to a source file, save it, hit Data → Refresh All, watch the dashboard update. Say while you do it: *"That's why we built the pipeline upstream. The dashboard isn't a one-off report — it's a live view that stays current as the source data does."*

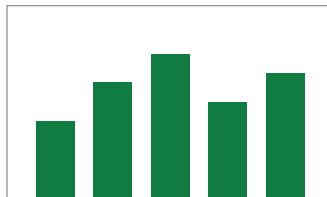
The Chart Test

The single biggest mistake in dashboards is the wrong chart for the question. Three questions to ask before you pick:

1. **Comparison, composition, or trend?** Comparing values across categories, showing parts of a whole, or tracking change over time?
2. **How many things?** 2–10 categories work for almost any chart. 50+ usually need a table, not a chart.
3. **Is time the X-axis?** If yes, line chart almost always. If no, bar or column.

COMPARISON

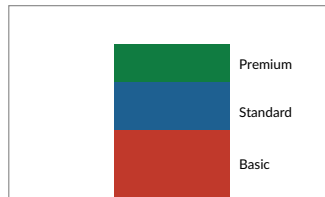
use BAR or COLUMN



"Hours by region"

COMPOSITION

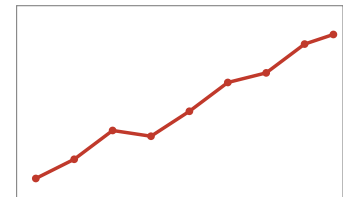
use STACKED BAR or PIE (≤ 5)



"Plan mix"

TREND

use LINE



"Sessions over time"

A note on pie charts. Pie charts only work for ≤ 5 slices, with clearly different sizes, when the parts add up to a meaningful whole. Beyond five slices, the human eye can't compare them properly. Reach for a stacked bar instead.

ASSESSOR FAVOURITE

Asked "why this chart, not another?", walk through the chart test out loud. "It's a comparison across five categories with no time element, so a column chart is the right choice. A pie would technically work but bars are easier to compare at a glance."

Halfway House

Four scenarios. Pick the right chart for each. Answers below the line.

1. Priya wants to see total watch hours for each of the 11 UK regions, in one view.

☐ **Pie** ☐ **Column / Bar** ☐ **Line** ☐ **Stacked bar**

2. She wants to see how monthly sessions changed across the 12 months of FY 2025-26.

☐ **Pie** ☐ **Column / Bar** ☐ **Line** ☐ **Stacked bar**

3. She wants to see what proportion of subscribers are Basic / Standard / Premium.

☐ **Pie** ☐ **Column / Bar** ☐ **Line** ☐ **Stacked bar**

4. She wants to compare, for each of 11 regions, the split of sessions across Basic / Standard / Premium plans.

☐ **Pie** ☐ **Column / Bar** ☐ **Line** ☐ **Stacked bar**

Answers – check yourself

1. **Column / Bar**. Comparison across 11 categories, no time axis. Pure column-chart territory.

2. **Line**. Time on the X-axis → line, almost always.

3. **Pie** – just about. Three slices, parts of a whole. A stacked bar would also work and is often clearer. Avoid pie if you can't tell which slice is biggest at a glance.

4. **Stacked bar**. Comparison *and* composition combined – exactly what stacked bar is for. One bar per region, segments coloured by plan.

Say it out loud

Three questions about the pivots and charts you just built. Practise them out loud now — the answers are short, but the assessor wants to hear that every choice was deliberate.

Q1 — Why this PivotTable layout, not another?

The assessor will pick one of your pivots and ask. Your answer names the row/column/values choices and what question each one answers. *"I put region in Rows and plan_type in Columns because Priya wants to see plans side by side per region. I put Count of record_id in Values because the question was 'how many sessions'. If she wanted total minutes I'd swap Count for Sum of duration."*

Q2 — Why this chart, not another?

The Chart Test, narrated. *"It's a comparison across regions, so column or bar — and there's no time element. Eleven regions is fine for a column chart, would have been too many for a pie. Pie charts only work for ≤ 5 slices anyway."* Pick one of your charts and rehearse this answer.

Q3 — Why Power Query rather than just Excel?

The one-sentence answer is about Refresh. *"Because next month, when new source data arrives, I overwrite the source file and click Data → Refresh All. The whole pipeline reruns — cleaning, merging, formulas, pivots, charts — in under thirty seconds. If I'd done this in Excel formulas alone, that would be half a day of rework, probably with a bug or two introduced."*



DISTINCTION TELL

For every PivotTable, be ready to justify three things: *what's in Rows and why, what's in Values and why, what aggregation (Sum, Count, Average) and why.* Three sentences. The pivot looks identical to anyone else's — but the assessor can hear in your answer that you chose every part of it deliberately.

Bullet Points — the chapter in 30 seconds

- **PivotTables** answer cross-tab questions in three drag-and-drops.
- **Slicers** make the dashboard interactive. Connect one slicer to multiple pivots and one click filters them all together.
- **PivotCharts** auto-update with their pivot.
- **The Chart Test** — comparison, composition, or trend? How many things? Time-based? — picks the right chart every time.
- **Refresh is the killer feature.** Data → Refresh All reruns the whole pipeline when source data changes.

TEST DRIVE

Before closing the laptop, run this self-check. Tick each box only when it's true:

- ☐ At least two PivotTables built on the Activity Table — different questions answered.
- ☐ At least one PivotChart linked to one of them.
- ☐ Slicers added (region, plan_type) and connected to multiple pivots if you have more than one.
- ☐ You ran the **Two-Route Check** on at least one pivot grand-total against a SUMIFS/COUNTIFS on the raw table.
- ☐ You can name the three questions in The Chart Test from memory.
- ☐ You demonstrated Refresh — pressed Data → Refresh All and watched the workbook update.
- ☐ You can answer "why Power Query, not just Excel?" in one sentence about Refresh.

COMING UP — DAY 7

Dashboard Polish. Tomorrow we turn today's pivots and charts into something that actually *looks* like a dashboard. Number formatting, Merge & Centre for KPI tiles, conditional formatting in depth, slicer styling, chart polish (removing the four most common pieces of chart junk), bucket grouping in PivotTables, and the named device — *The Polish Pyramid* — for laying it all out on a single sheet.

that's Day 6 done — the analytical layer is live.